

Task 8

Normalization database using functional dependencies upto BCNF

Aim:

(10) Normalize database using functional dependencies upto BCNF

Hospital Database

1. Identify - hospital attributes:

patient-ID, patient-Name, Doctor-ID, Doctor-Name, Department, Room-No, Treatment, Bill Amount

2. Define Relational Schema.

Hospital (patient-ID, patient-Name, Doctor-ID, Doctor-Name, Department, Room-No, Treatment, Bill Amount)

Step 2: Convert to 1NF

1. Eliminate repeating groups or arrays
2. Create separate tables for

Step 3: Convert to 2NF

1. Ensure Each non-key attribute depends on the entire primary key
 2. move non-key attribute to separate table if they depend only part of the primary key.
- Create Doctor table: Doctor (Doctor-ID, Doctor-Name, Department)
 - Create patient table: patient (patient-ID, patient-Name, Doctor-ID, Room-No, Treatment, Bill Amount)

Step 4: Convert to 3NF

1. Ensure there are no transitive dependencies
 2. move non-key attributes to separate tables if they depend on another non-key attributes
- Create room table: Room (Room-No, Department)
 - Update Doctor table: Doctor (Doctor-ID, Doctor-Name)

Step 4: Convert to 1A or

1. Ensure every determinant is a candidate key
2. Check for Overlapping Candidate key
3. Decompose Notation to Eliminate redundancy
4. No further decomposition needed

Using griffith tool:

1. Input relation Schema and functional dependencies
2. Griffith tool generates a dependency graph
3. Analyze the graph to identify normalization issues
4. Apply normalization rules to transform the Schema
5. Verify the resulting Schema meets BCNF Criteria

Griffith tool Steps:

1. Create a new project in griffith
2. Define the relational Schema and FD's
3. Run the Dependency graph tool
4. Analyze the graph for normalization tool
5. Apply Transformation using the "Apobiste" tool

Normalization Schema:

1. Patient (Patient-ID, Patient-Name, Doctor-ID)
2. Doctor (Doctor-ID, Doctor-Name)
3. Room (Room-No, Department)

VEL TECH - CSE	
EX NO.	08
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	-
TOTAL (20)	15
DATE	15/10/20

15/10/20

Result:

Thus, the Normalized database using functional dependencies upto using functional dependencies upto BCNF created successfully.